

1 Proposed Solution

This section details about the overall scope of this RFP project, including the overview of proposed solution.

1.1 Detail BOM (Bill of Material)

1.1.1 ANNEX-A1: COMPUTING NODES BoQ

A) Rack Mount Server (without GPU) QTY: 3

Part/ Model No.	Config# Server SYS-222C-TN without GPU	Unit QTY
SYS-222C-TN	X14DBHM + CSE-DC201, CloudDC DC-MHS FLW DP Intel 2U 24 B	1
EWCS	3 YRS LABOR, 3 YRS PARTS, CRS UNDER LIMITED WRNTY	1
P4X-GNR6747P-SRVEZ-XCC	GNR-SP 6747P 2P 48C(2.7/3.1G,330W)288M S512 T2048 4444 MRDIM	2
MEM-DR564MC-ER64	64GB DDR5 RDIMM 6400 ECC REG	24
HDS-M2N4-960G0-E1-TXD-NON-110	SSD M.2 NVMe PCIe4 960GB 1DWPDL TLC D, 110mm---HDS-M2N4-960G0-E1-TXD-NON-110	2
HDS-25N4-001T9-E1-TXD-NON-015	SSD 2.5" NVMe PCIe4 1.9TB 1DWPDL TLC D, 15mm---HDS-25N4-001T9-E1-TXD-NON-015	4
AOC-A25G-i2SG-P	AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P	1
AOC-LPE35002-M2	2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2	1
AOM-TPM-9672SCM-P	TPM 2.0 with Infineon SLB 9672 For SCM module,HF,RoHS---AOM-TPM-9672SCM-P	1
AOC-VROCPREMOD	Premium	1
PWS-2K09A-1R	AC-DC 1U 2000W 80 Plus Titanium CRPS, forward airflow, W73.5	2
SFT-DCMS-SINGLE	DataCenter Management Package (per node license)	1
BPN-NVME5-HS219N-S8	2U 8-port Hybrid backplane support up to 8xSAS3/SATA2/NV	1
MCP-240-20111-0N	Riser bracket I for SCDC201, short,RoHS	1
LBL-0968	Pack of 4 NVMe Labels for Gen 3, 2.5" HDD Tray (SC973)	2
RSC-M2R-6G5	2U RHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	2
RSC-M2-6G5	2U LHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	1
MCP-310-20109-0B	Mylar air shroud for SCDC201,RoHS	1
CBL-KIT-222C-TN-8	8-NVMe Cable Kit for X14 CloudDC SYS-222C-TN,RoHS	1
SNK-P0088P	2U Passive CPU HS for X13 EGS and X14-SP BHS-SP Systems	2
CBL-ASMC-1226MT16P	MCIOx16/RSC to Multi-trak x16+PWR,26cm,30#,RoHS	2
MCP-240-20110-0N	Riser bracket r for SCDC201, short,RoHS	2
MCP-220-00155-0B	2.5-inch New ID gen 3 tool-less (clip design) HDD tray wi/lo	8
SKT-1543H-0000-FXC	SOCKET E2 E2A CARRIER,LGA4710,GNR-SP XCC,DG1.0,COOL GRAY,HF	2
MC0037	Assembly	1
AOM-SFP28-25GbE-SR-1-MLN	MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR	2



B) Rack Mount server QTY: 1 with GPU Card I NVIDIA-L4 Card

Part/ Model No.	Config# Server SYS-222C-TN with 3x L4 GPU	QTY
SYS-222C-TN	X14DBHM + CSE-DC201, CloudDC DC-MHS FLW DP Intel 2U 24 B	1
EWCS	3 YRS LABOR, 3 YRS PARTS, CRS UNDER LIMITED WRNTY	1
P4X-GNR6747P-SRVEZ-XCC	GNR-SP 6747P 2P 48C(2.7/3.1G,330W)288M S512 T2048 4444 MRDIM	2
MEM-DR564MC-ER64	64GB DDR5 RDIMM 6400 ECC REG	24
HDS-M2N4-960G0-E1-TXD-NON-110	SSD M.2 NVMe PCIe4 960GB 1DWPD TLC D, 110mm---HDS-M2N4-960G0-E1-TXD-NON-110	2
HDS-25N4-001T9-E1-TXD-NON-015	SSD 2.5" NVMe PCIe4 1.9TB 1DWPD TLC D, 15mm---HDS-25N4-001T9-E1-TXD-NON-015	4
AOC-A25G-i2SG-P	AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P	2
GPU-NVL4	NVIDIA Ada L4 24GB GDDR6 PCIe Gen 4th-- 70w---GPU-NVL4	3
AOC-LPE35002-M2	2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2	2
AOM-TPM-9672SCM-P	TPM 2.0 with Infineon SLB 9672 For SCM module,HF,RoHS---AOM-TPM-9672SCM-P	1
AOC-VROCPREMOD	Premium	1
PWS-2K09A-1R	AC-DC 1U 2000W 80 Plus Titanium CRPS, forward airflow, W73.5	2
SFT-DCMS-SINGLE	DataCenter Management Package (per node license)	1
BPN-NVME5-HS219N-S8	2U 8-port Hybrid backplane support up to 8xSAS3/SATA2/NV	1
MCP-240-20111-0N	Riser bracket I for SCDC201, short,RoHS	1
LBL-0968	Pack of 4 NVMe Labels for Gen 3, 2.5" HDD Tray (SC973)	2
RSC-M2R-6G5	2U RHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	2
RSC-M2-6G5	2U LHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	1
MCP-310-20109-0B	Mylar air shroud for SCDC201,RoHS	1
CBL-KIT-222C-TN-8	8-NVMe Cable Kit for X14 CloudDC SYS-222C-TN,RoHS	1
CBL-ASMC-1226MT16P	MCIOx16/RSC to Multi-trak x16+PWR,26cm,30#,RoHS	2
MCP-240-20110-0N	Riser bracket r for SCDC201, short,RoHS	2
MCP-220-00155-0B	2.5-inch New ID gen 3 tool-less (clip design) HDD tray wi/lo	8
SKT-1543H-0000-FXC	SOCKET E2 E2A CARRIER,LGA4710,GNR-SP XCC,DG1.0,COOL GRAY,HF	2
FAN-0239L4	80x80x38 mm, 16.8K RPM, Middle Cooling Fan for 2U Hyper-S Se	4
SNK-P0087V	1U Passive CPU VC HS for X13 Intel Eagle Stream Platform	2
MC0037	Assembly	1
AOM-SFP28-25GbE-SR-1-MLN	MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR	4
NVIDIA GPU License	NVIDIA AI Enterprise Essentials Subscription per GPU	3



B) Rack Mount server QTY: 1 with GPU II NVIDIA-L40S Card

Part/ Model No.	Config# Server SYS-222C-TN with 2x L40s GPU	QTY
SYS-222C-TN	X14DBHM + CSE-DC201, CloudDC DC-MHS FLW DP Intel 2U 24 B	1
EWCS	0% 3 YRS LABOR, 3 YRS PARTS, 1 YR CRS UNDER LIMITED WRNTY	1
P4X-GNR6747P-SRVEZ-XCC	GNR-SP 6747P 2P 48C(2.7/3.1G,330W)288M S512 T2048 4444 MRDIM	2
MEM-DR564MC-ER64	64GB DDR5 RDIMM 6400 ECC REG	24
HDS-M2N4-960G0-E1-TXD-NON-110	SSD M.2 NVMe PCIe4 960GB 1DWP TLC D, 110mm---HDS-M2N4-960G0-E1-TXD-NON-110	2
HDS-25N4-001T9-E1-TXD-NON-015	SSD 2.5" NVMe PCIe4 1.9TB 1DWP TLC D, 15mm---HDS-25N4-001T9-E1-TXD-NON-015	4
AOC-A25G-i2SG-P	AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P	2
GPU-NVL40S	NVIDIA Ada L40S 48GB GDDR6 PCIe Gen 4th---GPU-NVL40S	2
AOC-LPE35002-M2	2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2	2
AOM-TPM-9672SCM-P	TPM 2.0 with Infineon SLB 9672 For SCM module,HF,RoHS---AOM-TPM-9672SCM-P	1
AOC-VROCPREMOD	Premium	1
PWS-2K09A-1R	AC-DC 1U 2000W 80 Plus Titanium CRPS, forward airflow, W73.5	2
SFT-DCMS-SINGLE	DataCenter Management Package (per node license)	1
MCP-240-20103-0N	Riser bracket r for SCDC201,RoHS	2
BPN-NVME5-HS219N-S8	2U 8-port Hybrid backplane support up to 8xSAS3/SATA2/NV	2
LBL-0968	Pack of 4 NVMe Labels for Gen 3, 2.5" HDD Tray (SC973)	4
MCP-240-20104-0N	Riser bracket l for SCDC201,RoHS	1
FAN-0239L4	80x80x38 mm, 16.8K RPM, Middle Cooling Fan for 2U Hyper-S Se	4
RSC-M2R-6G5	2U RHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	2
CBL-ASMC-1226MT16P	MCIOx16/RSC to Multi-trak x16+PWR,26cm,30#,RoHS	2
RSC-M2-6G5	2U LHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	1
MCP-220-00155-0B	2.5-inch New ID gen 3 tool-less (clip design) HDD tray wi/lo	16
CBL-KIT-222C-TN-16	16-NVMe Cable Kit for X14 CloudDC SYS-222C-TN,RoHS	1
MCP-310-82619-0B	Sponge air Block sets for CSE-826B and LA26-AW,RoHS	1
CBL-PWEX-1364-26	GPU Power Cable - 12VHPWR 2x6+4S to 2x4,MicroHi,26cm,6.25A/p,16AWG,RoHS	2
SKT-1543H-0000-FXC	SOCKET E2 E2A CARRIER,LGA4710,GNR-SP XCC,DG1.0,COOL GRAY,HF	2
MCP-310-20117-0B	Mylar air shroud for SCDC201,for GPU config,RoHS	1
SNK-P0087V	1U Passive CPU VC HS for X13 Intel Eagle Stream Platform	2
MC0037	Assembly	1
AOM-SFP28-25GbE-SR-1-MLN	MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR	4
NVIDIA GPU License	NVIDIA AI Enterprise Essentials Subscription per GPU	02



B) Rack Mount server QTY: 1 with GPU Card III NVIDIA-H200 Card

Part/ Model No.	Config# Server SYS-222C-TN with 1x H200	QTY
SYS-222C-TN	X14DBHM + CSE-DC201, CloudDC DC-MHS FLW DP Intel 2U 24 B	1
EWCS	0% 3 YRS LABOR, 3 YRS PARTS, 1 YR CRS UNDER LIMITED WRNTY	1
P4X-GNR6740P-SRV5H-XCC	GNR-SP 6740P 2P 48C HT 2.1G 288M 270W SGX128 TDX1024 2222 7y	2
MEM-DR564MC-ER64	64GB DDR5 RDIMM 6400 ECC REG	24
HDS-M2N4-960G0-E1-TXD-NON-110	SSD M.2 NVMe PCIe4 960GB 1DWPDL TLC D, 110mm---HDS-M2N4-960G0-E1-TXD-NON-110	2
HDS-25N4-001T9-E1-TXD-NON-015	SSD 2.5" NVMe PCIe4 1.9TB 1DWPDL TLC D, 15mm---HDS-25N4-001T9-E1-TXD-NON-015	4
AOC-A25G-i2SG-P	AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P	2
GPU-NVH200NVL	NVIDIA H200 NVL 141GB PCIe 5.0 x16, Air Cool---GPU-NVH200NVL	1
AOC-LPE35002-M2	2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2	2
AOM-TPM-9672SCM-P	TPM 2.0 with Infineon SLB 9672 For SCM module,HF,RoHS---AOM-TPM-9672SCM-P	1
AOC-VROCPREMOD	Premium	1
PWS-3K21A-1R2	3200W, Titanium efficiency, Redundant w/PMBus power supply---PWS-3K21A-1R2	2
SFT-DCMS-SINGLE	DataCenter Management Package (per node license)	1
CBL-PWEX-1186Y-25	Fan,1X6/P2.54 to 2 2x4/P2.5 NanoFit, 25cm22/26AWG,5A/Pin	3
RSC-M2R-6G5	2U RHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	2
MCP-240-20121-0N	Riser bracket M Assy for DC201 (w/ BF3 & CX7),RoHS	1
RSC-M2-6G5	2U LHS DC-MHS Riser card with one PCIe 5.0 X16 slot,HF,R	1
CBL-PWEX-1364Y-30	12V-2x6+4S to 2X4Y,MicrHi,30cm,8.33A/p,16AWG,RoHS	2
CBL-ASMC-1226MT16P	MCIOx16/RSC to Multi-trak x16+PWR,26cm,30#,RoHS	1
MCP-240-20119-0N	Riser bracket R Assy for DC201 (w/ RTX PRO 6000),RoHS	1
MCP-240-20120-0N	Riser bracket L Assy for DC201 (w/ RTX PRO 6000),RoHS	1
MCP-220-00155-0B	2.5-inch New ID gen 3 tool-less (clip design) HDD tray wi/lo	8
BPN-NVME5-HS219N-S8	2U 8-port Hybrid backplane support up to 8xSAS3/SATA2/NV	1
CBL-KIT-222C-TN-8	8-NVMe Cable Kit for X14 CloudDC SYS-222C-TN,RoHS	1
FAN-0239L4	80x80x38 mm, 16.8K RPM, Middle Cooling Fan for 2U Hyper-S Se	4
LBL-0968	Pack of 4 NVMe Labels for Gen 3, 2.5" HDD Tray (SC973)	2
MCP-310-20121-0B	Mylar Air Shroud for DC201 (w/ RTX PRO 6000),RoHS	1
SNK-P0087V	1U Passive CPU VC HS for X13 Intel Eagle Stream Platform	2
SKT-1543H-0000-FXC	SOCKET E2 E2A CARRIER,LGA4710,GNR-SP XCC,DG1.0,COOL GRAY,HF	2
CBL-PWCD-0925	PWCD,US,NEMA 5-15P TO IEC60320 C19, 6FT, 14AWG, 15A/125V	2
MC0037	Assembly	1
AOM-SFP28-25GbE-SR-1-MLN	MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR	4



1.1.2 ANNEX-A2: STORAGES BoQ

Primary Storages with RAW storage capacity of 350 TB QTY: 1

No.	Part Number	Model	Description	Unit Qty.	Qty.
1	OceanStor Dorado 6000				
1.1	OceanStor Dorado 6000 Main Equipment				
1.1.1	Dual-Controller Unit 1				
	Controller Enclosure				
	02357KTB	D6V7N-2C1T-O-E	OceanStor Dorado 6000 (2U,Dual Ctrl,NVMe,AC\240V HVDC,1TB Cache,4*100Gb RDMA,36*Palm,SPE64C0236,V7,Applicable to 1.2m-Depth Cabinets,C13)	1	1
	Expanding Interface Module				
	03050RMD	SMART32GbFE	4 ports DPU-Based SmartIO I/O module(SFP28,32Gb FC)	2	2
	03050RMG	V7S4*10GE	4 ports DPU-Based SmartIO I/O module(SFP+,10Gb ETH)	2	2
	Disk Components				
	02552383	D6V7-NVMe15.36-E	15.36TB SSD NVMe Palm Encryption Disk Unit(7")	24	24
1.2	Installation Material				
	14130858	SN2F01FCPC	Patch Cord,DLC/PC,DLC/PC,Multi-mode,3m,A1a.2,2mm,42mm DLC,OM3 bending insensitive	16	16
1.3	Software				
	8803011267	D6V7-LBS-Super	SAN License Bundle (Including Thin,Snap,Replication,Clone,QoS,Erase,DME IQ,Virtualization,Metro,CDP,Multi-Tenant)	1	1
	8803011271	D6V7-LBS-Super-NAS	NAS License Bundle (Including CIFS,NFS,NDMP,Thin,Snap,Replication,Clone,QoS,Erase,Metro,CDP,Lock,DME IQ,Audit,DNS,Move)	1	1
	8803011289	D6V7-DDCM-C-SUP-RES2	SmartDedupe & SmartCompression Software License General Edition	1	1



Warranty and Support Services

No.	Part Number	Model	Description	Unit Qty.	Qty.
	Storage-OceanStor Dorado 6000 V700R001_Site1				
2	OceanStor Dorado 6000				1
	88134UHK-2A6	8803011267_8 8134UHK-2A6_36	SAN License Bundle (Including Thin,Snap,Replication,Clone,QoS,Erase,DME IQ,Virtualization,Metro,CDP,Multi-Tenant)_Hi-Care Application Software Upgrade Support Service Dorado 6000 SAN License Bundle_36Month(s)	1	1
	88134UHK-2A4	8803011271_8 8134UHK-2A4_36	NAS License Bundle (Including CIFS,NFS,NDMP,Thin,Snap,Replication,Clone,QoS,Erase,Metro,CDP,Lock,DME IQ,Audit,DNS,Move)_Hi-Care Application Software Upgrade Support Service Dorado 6000 NAS License Bundle_36Month(s)	1	1

OceanStor Dorado 6000 Storage IOPS:

New-Gen OceanStor Dorado 6000 Converged All-Flash Storage



350 TB
RAW Capacity

788471
IOPS

Parameters	Specification
Controller	1TB
Front-end I/O module	8x32G FC,8x10G Optics
Disk	24x15.36TB NVMe SSD(7,Encryption Disk)
Value-added	////
E2E reduction ratio	1.0

Converged

- SAN-NAS-S3 native convergence reduces TCO by 25%.
- FlashEver cross-gen convergence eliminates host service interruptions during storage replacement.

Ever Fast

- FlashLink® intelligent algorithm with data-control plane separation improves performance by 50%.
- LUN active-active architecture and the scale-out file system balance performance across multiple controllers.

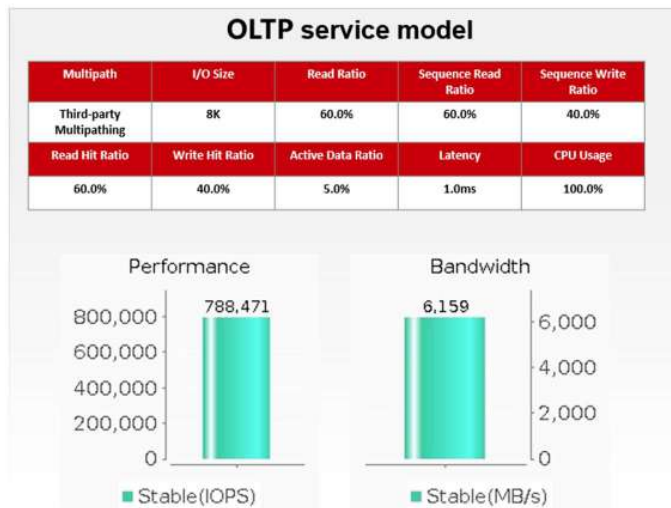
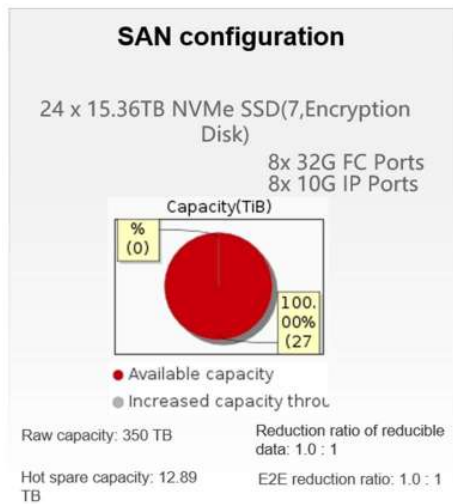
Ever Solid

- The industry's only gateway-free SAN-NAS-S3 active-active solution achieves zero RPO and near-zero RTO for mission-critical services.
- Built-in ransomware detection for blocks and files achieves a detection rate of up to 99.99%.

1 Huawei Proprietary - Restricted Distribution



SAN: Delivers 350 TB RAW Capacity and 788471 IOPS in OLTP Scenarios



2 Huawei Proprietary - Restricted Distribution



Backup Storages with useable capacity of 50 TB QTY: 1

No.	Part Number	Model	Description	Unit Qty.	Qty.
1	OceanStor 5210				
1.1	OceanStor 5210 Main Equipment				
1.1.1	Controller Unit 1				
	Controller Enclosure				
	02356JVS	521V6-S-S-64G-35HW	5210(2U, Dual Controllers, SAS, AC\240V HVDC, 64GB Cache, 8*1Gb ETH,4*10Gb ETH(Including Multi-Mode SFP+),4*SAS3.0 port,12*3.5 Inch SAS,SPE26C0212)	1	1
	Expansion Interface Module				
	03050RUT	ODK-SIO4*32FC-ML	4 ports SmartIO I/O module(SFP28,32Gb FC)	2	2
	Disk Components				
	02354CQD	L1-L-NLSAS10TV5	10TB 7.2K RPM NL-SAS Disk Unit(3.5")	10	10
	02356TJW	L2-35-SSD960G	960GB SSD SAS Disk Unit(3.5")	6	6
	Disk Enclosure				
	02357BGF	DAE62435U4E V5	SAS Disk Enclosure(4U,AC\240V HVDC,3.5 Inch,Expansion Module,24 Disk Slots,Without Disk Units,DAE62435U4)	1	1
1.2	Storage Accessories Package				
1.2.1	Optical Fiber				
	14130858	SN2F01FCPC	Patch Cord,DLC/PC,DLC/PC,Multi-mode,3m,A1a.2,2mm,42mm DLC,OM3 bending insensitive	12	12
1.2.2	PANGEA V6 SAS Cable				
	04052252	HS-SAS-3-02	High Speed Cable,48G Mini SAS HD Cable,3m,SFF 8644 Plug,28AWG*4P*2B(S),SFF 8644 Plug,indoor,SAS 3.0,LSZH,EEPROM	2	2
1.3	System Software				



	88038NDA	LIC-5210V6-AP	SAN&NAS Advanced Package(Including DeviceManager,DME IQ,Acceleration,CIFS,NFS,Audit,DNS,Thin,Migration(SAN),Snap,Clone,QoS,Erase,CDP,UltraPath,WORM,Multi-Tenant,Quota,Vault,Virtualization,Metro,Replication)	1	1
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Warranty and Support Services

No.	Part Number	Model	Description	Unit Qty.	Qty.
			Storage Equipment-OceanStor 5210 V700R001_Site1		
2			OceanStor 5210		1
	88134ULC-349	02356JVS_8 8134ULC-349_36	5210(2U, Dual Controllers, SAS, AC\240V HVDC, 64GB Cache, 8*1Gb ETH,4*10Gb ETH(Including Multi-Mode SFP+),4*SAS3.0 port,12*3.5 Inch SAS,SPE26C0212)_Hi-Care Onsite Standard OceanStor 5210 V5 Controller Enclosure_36Month(s)	1	1
	88134ULC-240	02356TJW_8 88134ULC-240_36	960GB SSD SAS Disk Unit(3.5")_Hi-Care Onsite Standard OceanStor 2200/2600/51xx/52xx 960GB SSD_36Month(s)	6	6
	88134ULC-345	02357BGF_8 8134ULC-345_36	SAS Disk Enclosure(4U,AC\240V HVDC,3.5 Inch,Expansion Module,24 Disk Slots,Without Disk Units,DAE62435U4)_Hi-Care Onsite Standard OceanStor 5110/5210 V5 Disk Enclosure_36Month(s)	1	1
	88134UHK-25M	88038NDA_88134UHK-25M_36	SAN&NAS Advanced Package(Including DeviceManager,DME IQ,Acceleration,CIFS,NFS,Audit,DNS,Thin,Migration(SAN),Snap,Clone,QoS,Erase,CDP,UltraPath,WORM,Multi-Tenant,Quota,Vault,Virtualization,Metro,Replication)_Hi-Care Application Software Upgrade Support Service OceanStor 5210 SAN&NAS Advanced Package_36Month(s)	1	1



OceanStor 5210 Storage IOPS:



SAN switches QTY: 2

Part number	Product Description	Qty
7D5JCTO1WW	Switch : Lenovo ThinkSystem DB720S, 24 ports active with 32Gb SWL SFP's, 2PS (Port side exhaust), rail kit, 3Yr FW	2
BF60	Lenovo ThinkSystem DB720S, 24 Ports Active with 32Gb SWL SFPs, 2PS (Port side exhaust), Rail Kit, 3Yr FW	2
ASR8	Lenovo 5m LC-LC OM3 MMF Cable	48
6201	1.5m, 10A/100-250V, C13 to C14 Jumper Cord	4
7Q02CTS2WW	STORAGE PREMIER NBD RESP	2
QA0Y	Months	72
QA11	Next Business Day	2
QA18	Premier	2

1.1.3 ANNEX-A3: SOFTWARE AND TRAINING BoQ

Software (Virtualization, Cloud) NEXAVM Software and Service QTY: 1 Job

LINE ITEM	Product Code	Product Type	Product Name	DESCRIPTION	QTY
1-1	NEXAVM Software and service	Software	nCSSV	CLOUD SERVER VIRTUALIZATION, HA, DRS, Automated Hot Add, backup, clone,.... with microsegmentation, VPC, vRouter, multitenant, management of baremetal, VMware clusters. Licensed per physical CPU socket	12
1-2		Software	nSDS	Distributed converged storage, support block, file and object storages, 2-6 copies or EC 4+2/EC8+2, SD read and write acceleration, Striping, support block, file and object storage, stretched cluster and Dedup and compression. 1TB License for raw data disk.	60
1-3		Software	nBDR	Perpetual License - Per CPU Socket - Enterprise Edition - VM Backup Module	12
1-4		Software	nKU	Container Management licensed per vCPU	32
1-5		Support	nSUS-1	Software Upgrade Services - 1 year	3
1-6		Support	nTSS-1	24*7 Remote Technical Support Services - 1 year	3

Training QTY: 1 Job

No.	Part Number	Model	Description	Qty.	Category
1	OceanStor Storage Deployment and Management Training_Local Training				
	88991XBX	88991XBX	OceanStor Storage Deployment and Management Training_Local Training_Quote by Class_3.0Day(s)_10Person(s)_1Class(es)	1	Training and Certification Service
	88991XBX	88991XBX	OceanStor Dorado Storage Advanced O&M Training_Local Training_Quote by Class_2.0Day(s)_10Person(s)_1Class(es)	1	Training and Certification Service



No.	Part Number	Model	Description	Qty.
1	LS			
1.1	Huawei Career Certification Exam Voucher			
	88970715	88970715	Storage Certification Voucher- HCIP_Person_10	10

2 Product & Software Detail

S. No.	Noun	Description	Attachment
1	A) Rack Mount Server	Supermicro SYS-222C-TN	Datasheet attached
2	B) Rack Mount server with GPU	Supermicro SYS-222C-TN	Datasheet attached
3	Primary Storage	Huawei OceanStor Dorado 6000	Datasheet/ Supporting Documents attached
4	Backup Storage	Huawei OceanStor 5210	Datasheet/ Supporting Documents attached
5	SAN switch	Lenovo ThinkSystem DB720S	Datasheet is attached
6	SOFTWARE AND TRAINING	NexaVM Cloud Feature List, nCSSV and nBDR	Datasheet/ Supporting Documents attached
7	COMMISSIONING AND SLA	ANNEX-C SERVICE LEVEL AGREEMENT	Document attached



3 Statement of Services

3.1 Statement of Work (SOW)

Techaccess Pakistan will deliver the professional services, as described in this Statement of Work ("SOW"), to ("NTC"). Techaccess will deliver only those services as expressly described further in this document and under the assumptions, dependencies, and conditions as described further in this document.

The Parties acknowledge that the precise scope of the services shall be clarified as a result of the first stages of each Service, and that the proposal, scope and pricing may have to be modified accordingly, if required by making use of the change control or escalation process defined herein. This Statement of Work is based on information currently available at Techaccess disposal as provided by NTC name Pakistan. Should any of this information be incorrect or incomplete or should material information be unknown to Techaccess, then Techaccess may modify its proposal or the services, or initiate the change control or escalation process defined herein.

This document contains the following sections:

- **Summary of Services**
Describes the services that Techaccess would be performing to implement the proposed solution
- **General**
Provides general assumption, pre-requisites, customer's responsibilities, exclusions, change request process etc.

4 Summary of Services

A summary of professional services to be provided under this project is outlined below:

- Provision of Supermicro SYS-222C-TN, Huawei Dorado 6000 Storage, Backup Storage OceanStor 5210 and SAN Switches Lenovo ThinkSystem DB720S as per the approved BOQ
- Complete setup, configuration, and integration of storage and servers and SAN switches.
- Comprehensive performance, redundancy, and security testing to ensure system stability and compliance.
- Delivery of detailed implementation documents, user training, and formal project handover upon successful completion.



5 Compliance Sheet

5.1 ANNEX-B: TECHNICAL SPECIFICATION COMPLIANCE

ANNEX-B1: COMPUTING NODES							
S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/N C)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
A Server without GPU Cards, Quoted Model: Supermicro SYS-222C-TN							
1	Processor	Rack Mount Chassis (Server Type A: with GPU capability only) having 2x Intel Xeon 6th Gen Processors, performance Cores >=48 (per Processor), >=2.4 GHz base frequency, cache >=48 MB.	M	FC	As discussed with customer without GPU Capability Server Type A required so, quoted SYS-222C-TN Server Type A is without GPU Capability, 2x Intel® Xeon GNR-SP 6747P 2P 48C(2.7/3.1G,330W)2 88M S512 T2048 4444 MRDIM		
2	Processor Preference	Higher Processor specs / Cores capacity will be preferred	10	FC	Scalable Higher end Processor refer to Datasheet	10	10
3	Memory (RAM)	RAM DDR5 >= 1.5TB per node with >= 64GB modules (>=6400 MT/s Memory speed).	M	FC	24x 64GB DDR5 RDIMM 6400 ECC REG		
4	Memory (RAM) Preference	Higher Memory speed / Capacity will be preferred	10	FC	For above calculation TruDDR5 RAM: 24x 64GB= 1536 GB = 1.536TB per server node.	10	10
5	Storage	2x Disk Drivers for OS/Hypervisor & >= 4 Disk Drivers NVME SSD >=4TB usable capacity per Node/server machine	M	FC	2x SSD M.2 NVMe PCIe4 960GB 1DWPD TLC D and 4x SSD 2.5" NVMe PCIe4 1.9TB 1DWPD TLC D		
6	Storage Preference	Higher Disk speed/capacity will be preferred	10	FC	For above calculation OS/Hypervisor Disk: 2x 960GB NVMe SSD Disk Drives: 4x 1.9TB NVMe SSD	10	10



7	Network/Interfaces	2x Port 25 GbE/each port NICs Per server node 2x Port 32 Gb / each port Fibre Channel HBA NICs per server node Dual Server Management ports per server node All required cables, transceivers/SFPs, and accessories need to be included	M	FC	AIOM G4 x8 2x 25GbE SFP28 Intel E810- XXVAM2,RoHS---AOC- A25G-i2SG-P with 2x MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,L C-LC, 850nm, SR, 2- ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8--- AOC-LPE35002-M2 with required cables		
8	Networking Preference	Higher Interface speed/Quantity of interfaces will be preferred.	10	FC	LAN: 2 RJ45 LAN port(s) (via DC-SCM)	10	10
					Obtained Marks =	40	40

ANNEX-B1: COMPUTING NODES

S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/NC)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
B	Server with GPU Card I NVIDIA-L4, Quoted Model: Supermicro SYS-222C-TN with NVIDIA Ada L4 24GB GPU Card						
9	Processor	Rack Mount Chassis (Type B with GPU compatibility and GPU cards loading) having 2x Intel Xeon 6th Gen Processors, performance Cores >=48 (per Processor), >=2.4 GHz base frequency, cache >=48 MB.	M	FC	Quoted SYS-222C-TN with GPU Card: NVIDIA Ada L4 24GB Server Type B with GPU compatibility, 2x Intel® Xeon GNR-SP 6747P 2P 48C(2.7/3.1G,330W)288 M S512 T2048 4444 MRDIM		
10	Processor Preference	Higher Processor specs / Cores capacity will be preferred	10	FC	Scalable Higher end Processor refer to Datasheet	10	10
11	Memory (RAM)	RAM DDR5 >= 1.5 TB per node with >= 64GB modules (>=6400 MT/s Memory speed).	M	FC	24x 64GB DDR5 RDIMM 6400 ECC REG		
12	Memory (RAM) Preference	Higher Memory speed / Capacity will be preferred	10	FC	For above calculation TruDDR5 RAM: 24x 64GB= 1536 GB = 1.536TB per server node.	10	



13	Storage	2x NVMe SSD Drives ≥600 GB (for OS/Hypervisor) & ≥ 4 Disk Drivers NVMe SSD ≥4TB usable capacity per Node/server	M	FC	2x SSD M.2 NVMe PCIe4 960GB 1DWPD TLC D and 4x SSD 2.5" NVMe PCIe4 1.9TB 1DWPD TLC D		
14	Storage Preference	Higher Disk speed/capacity will be preferred	10	FC	For above calculation OS/Hypervisor Disk: 2x 960GB NVMe SSD Disk Drives: 4x 1.9TB NVMe SSD	10	10
15	Networking	2x Dual Ports (4 ports total) 25 GbE/each port NICs Per server node 2x Dual Port (4 ports total) 32 Gb / each port Fibre Channel HBA NICs per server node Dual Server Management ports per server node All required cables, transceivers, SFP and accessories need to be included	M	FC	2x AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P with 4x MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR, 2x 2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2 with required cables		
16	Networking Preference	Higher Interface speed/Quantity of interfaces will be preferred.	10	FC	LAN: 2 RJ45 LAN port(s) (via DC-SCM)	10	10
17	GPU Card I	NVIDIA-L4 for or equivalent/higher GPU cards Each nodes with 1x NVIDIA L4 GPU or equivalent including necessary licenses for GPU virtualization to enable NTC to provide GPU based services in Multitenancy, OEM Support, license renewals and updating services for 03x years).	M	FC	3x NVIDIA Ada L4 24GB GDDR6 PCIe Gen 4th--70w---GPU-NVL4 NVIDIA with AI Enterprise (Essentials) Subscription		
Obtained Marks =						40	40



ANNEX-B1: COMPUTING NODES

S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/NC)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
B	Server with GPU II NVIDIA-L40S Card Quoted Model: Supermicro SYS-222C-TN with NVIDIA Ada L40S 48GB GPU Card						
9	Processor	Rack Mount Chassis (Type B with GPU compatibility and GPU cards loading) having 2x Intel Xeon 6th Gen Processors, performance Cores >=48 (per Processor), >=2.4 GHz base frequency, cache >=48 MB.	M	FC	Quoted SYS-222C-TN with GPU Card: NVIDIA Ada L40S 48GB Server Type B with GPU compatibility, 2x Intel® Xeon GNR-SP 6747P 2P 48C(2.7/3.1G,330W)288 M S512 T2048 4444 MRDIM		
10	Processor Preference	Higher Processor specs / Cores capacity will be preferred	10	FC	Scalable Higher end Processor refer to Datasheet	10	10
11	Memory (RAM)	RAM DDR5 >= 1.5 TB per node with >= 64GB modules (>=6400 MT/s Memory speed).	M	FC	24x 64GB DDR5 RDIMM 6400 ECC REG		
12	Memory (RAM) Preference	Higher Memory speed / Capacity will be preferred	10	FC	For above calculation TruDDR5 RAM: 24x 64GB= 1536 GB = 1.536TB per server node.	10	10
13	Storage	2x NVMe SSD Drives >=600 GB (for OS/Hypervisor) & >= 4 Disk Drivers NVMe SSD >=4TB usable capacity per Node/server	M	FC	2x SSD M.2 NVMe PCIe4 960GB 1DWPD TLC D and 4x SSD 2.5" NVMe PCIe4 1.9TB 1DWPD TLC D		
14	Storage Preference	Higher Disk speed/capacity will be preferred	10	FC	For above calculation OS/Hypervisor Disk: 2x 960GB NVMe SSD Disk Drives: 4x 1.9TB NVMe SSD	10	10
15	Networking	2x Dual Ports (4 ports total) 25 GbE/each port NICs Per server node 2x Dual Port (4 ports total) 32 Gb / each port Fibre Channel HBA NICs per server node Dual Server Management ports per server node All required cables,	M	FC	2x AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P with 4x MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR, 2x 2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-		



		transceivers, SFP and accessories need to be included			LPE35002-M2 with required cables		
16	Networking Preference	Higher Interface speed/Quantity of interfaces will be preferred.	10	FC	LAN: 2 RJ45 LAN port(s) (via DC-SCM)	10	10
18	GPU Card II	NVIDIA-L40S or equivalent/higher GPU cards including necessary licenses for GPU virtualization (other applicable licenses) to enable NTC to provide GPU based services in Multitenancy (if applicable), OEM Support, license renewals and updating services for 03x years)	M	FC	2x NVIDIA Ada L40S 48GB GDDR6 PCIe Gen 4th---GPU-NVL40S with AI Enterprise (Essentials) Subscription		
Obtained Marks =						40	40

ANNEX-B1: COMPUTING NODES

S#	Items	Required Item Description	Comp liance	Bidder Compli ance (FC/PC /NC)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
B	Server with GPU Card III NVIDIA-H200 Card Quoted Model: Supermicro SYS-222C-TN with NVIDIA H200 NVL 141GB GPU Card						
9	Processor	Rack Mount Chassis (Type B with GPU compatibility and GPU cards loading) having 2x Intel Xeon 6th Gen Processors, performance Cores >=48 (per Processor), >=2.4 GHz base frequency, cache >=48 MB.	M	FC	Quoted SYS-222C-TN with GPU Card: NVIDIA H200 NVL 141GB Server Type B with GPU compatibility, 2x Intel® Xeon GNR-SP 6740P 2P 48C HT 2.1G 288M 270W SGX128 TDX1024 2222 7y		
10	Processor Preference	Higher Processor specs / Cores capacity will be preferred	10	FC	Scalable Higher end Processor refer to Datasheet	10	10
11	Memory (RAM)	RAM DDR5 >= 1.5 TB per node with >= 64GB modules (>=6400 MT/s Memory speed).	M	FC	24x 64GB DDR5 RDIMM 6400 ECC REG		
12	Memory (RAM) Preference	Higher Memory speed / Capacity will be preferred	10	FC	For above calculation TruDDR5 RAM: 24x 64GB= 1536 GB =	10	10



					1.536TB per server node.		
13	Storage	2x NVMe SSD Drives >=600 GB (for OS/Hypervisor) & >= 4 Disk Drivers NVME SSD >=4TB usable capacity per Node/server	M	FC	2x SSD M.2 NVMe PCIe4 960GB 1DWPD TLC D and 4x SSD 2.5" NVMe PCIe4 1.9TB 1DWPD TLC D		
14	Storage Preference	Higher Disk speed/capacity will be preferred	10	FC	For above calculation OS/Hypervisor Disk: 2x 960GB NVMe SSD Disk Drives: 4x 1.9TB NVMe SSD	10	10
15	Networking	2x Dual Ports (4 ports total) 25 GbE/each port NICs Per server node 2x Dual Port (4 ports total) 32 Gb / each port Fibre Channel HBA NICs per server node Dual Server Management ports per server node All required cables, transceivers, SFP and accessories need to be included	M	FC	2x AIOM G4 x8 2x 25GbE SFP28 Intel E810-XXVAM2,RoHS---AOC-A25G-i2SG-P with 4x MMA2P00-AS, Mellanox TXCVR,25GbE,SFP28,LC-LC, 850nm, SR, 2x 2-ports HBA Emulex 32GFC/ 64GFC HBA woptics PCIe4 x8---AOC-LPE35002-M2 with required cables		
16	NetworkingP reference	Higher Interface speed/Quantity of interfaces will be preferred.	10	FC	LAN: 2 RJ45 LAN port(s) (via DC-SCM)	10	10
19	GPU Card III	NVIDIA-H200 or equivalent/higher GPU cards including necessary licenses for GPU virtualization (other applicable licenses) to enable NTC to provide GPU based services,(OEM Support, license renewals and updating services for 03x years)	M	FC	NVIDIA H200 NVL 141GB PCIe 5.0 x16, Air Cool---GPU-NVH200NVL with licence		
20	GPU Card IV	NVIDIA-A100 or equivalent/higher GPU cards including necessary licenses for GPU virtualization (MIG) to enable NTC to provide GPU based services in Multitenancy,(OEM Support, license renewals and updating services for 03x years)	M	NC	NVIDIA-A100 is End of Life no more available		



21	Deployment, Software, Support and Services	All necessary software, Firmware, Driver etc including management, monitoring and logging licenses for server nodes and GPU cards with one time deployment, commissioning & 3x years (post PAC) OEM support with license renewal and updation services	M	FC	One time deployment will be done by the Techaccess/Supermicro		
					Obtained Marks =	40	40

ANNEX-B2: NVMe STORAGE (Quantity = 01)							
S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/NC)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
1	Architecture:	Proposed Modular All Flash storage system must contain dual Active/Active Controller, based on CISC or RISC processor architecture.	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage Systems V7 Data Sheet.pdf Active-active: Page no. 5: Topic proposed solution new-gen storage also uses the SmartMatrix full-mesh technology with solid stability and the active-active setup that integrates block, file, and object to achieve high reliability.		
2		Intel/CISC processor based controllers	15			0	1
3		Minimum, 32 cores per controller and 64 cores overall	Mandatory	FC	Cores per controller: Reference to the supporting document Dorado 6000 V7 - CPU cores.pdf, the		

					proposed solution provides Kunpeng 920 7260 64-core 2.6 GHz x 2 (no longer deliverable since April 11, 2026) /Kunpeng 920 5250Z 48-core (96 threads) 2.6 GHz x 2 (deliverable since April 11, 2026) Enhanced version: Kunpeng 920 7260Z 64-core (128 threads) 2.6 GHz x 2		
4		Higher than 32 cores per controller	10	FC	Cores per controller: Reference to the supporting document Dorado 6000 V7 - CPU cores.pdf, the proposed solution provides Kunpeng 920 7260Z 64-core (128 threads) 2.6 GHz x 2, which meets the requirement of cores higher than 32 cores per controller.	10	10
5	Disk Type:	Should have NVME-TLC or better performance/ technology disk type as per industry standards.	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado NVMe SSD Data Sheet.pdf NVMe-TLC: Page no. 2: The proposed solution provides TLC Flash Chip Type NVMe.		
6	Storage Capacity:	Should provide at least 350 TB RAW capacity of SAN without deduplication and compression, with all requisite licenses.	Mandatory	FC	Reference to the Technical BOQ the proposed solution is offered with RAW Capacity(TB) = 363.21		
7		Per Disk capacity = 15TB	10	FC	Reference to the Technical BOQ the proposed solution is offered with 15.36 TB SSD NVMe disk capacity.	10	1



8	Hot-Spare Disk:	Should have global hot-spares disks as recommended by OEM (System need to work even in case of 2x disk drives failure with no impact on user capacity).	15	FC	Reference to the Technical BOQ the proposed solution offers Hot Spare Policy of 2 disks.	15	15
9	Cache:	At least 384 GB DRAM cache per controller, (Total minimum 768 GB for complete storage).	Mandatory	FC	Reference to the Technical BOQ the proposed solution provides 1 TB cache for complete storage.		
10		More than 384 GB DRAM cache per controller, (More than 768 GB for complete storage)	15	FC	Reference to the Technical BOQ the proposed solution provides 1 TB cache for complete storage, which meets the requirement of complete storage more than 768 GB.	15	15
11	RAID Configuration:	Proposed storages should support Distributed RAID/Dynamic RAID, RAID6 etc	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage Systems V7 Data Sheet.pdf RAID 6: Page no. 5: The proposed solution offers RAID 5, RAID 6, RAID 10, RAID-TP, cross-enclosure RAID* .		
12	Backend Connectivity:	End-to-end NVME Backend connectivity for disks drives	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf Back-end interconnection: Page no. 4: Topic The storage systems support 64 Gbit/s Fibre Channel, the proposed solution supports 100 Gbit/s		



					RDMA, and NVMe for front-end interconnection, cluster switching, and back-end interconnection , and can offload specific tasks to hardware to free CPU resources, ensuring low latency and high bandwidth.		
13	Frontend Ports:	Minimum 2 x 32Gbps FC Ports for FC connectivity per controller.	Mandatory	FC	Frontend Ports: Reference to the technical BOQ, the proposed storage solution provides 2 x (4 x 32 Gbps FC ports).		
14		Ethernet ports 4x 10 Gbps connectivity, 1 Gbps (Management port) and per controller	Mandatory	FC	Frontend Ports: Reference to the technical BOQ, the proposed storage solution provides 2 x (4 x 10 Gbps ETH ports).		
15	Protocol Support:	Should support FC, iSCSI, NFS, NVMe/TCP, NVMe/FC etc (Please specify all protocols)	5	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage Systems V7 Data Sheet.pdf Supported Storage Protocols: Page no. 5: The proposed solution supports storage protocols FC, FC-NVMe, iSCSI , NVMe over RoCE, NVMe over TCP , NFS , SMB, NFS over RDMA, S3.	5	5
16	High Availability:	Redundant disk enclosure connectivity with Redundant (front-end & back-end) interfaces on each controller, Redundant fans & power supplies for controllers & disk enclosures.	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf High Availability: Page no. 146 - 147: Topic 8.2.4 Array-level Redundancy , the		

					proposed solution provides intra-array high availability protection for services requiring high reliability, OceanStor Dorado provides array- level (site)active-active protection to ensure service continuity in case of a power failure or disaster such as earthquake and fire.		
17	Deduplicatio n / Compression:	Must have both in-line and post process deduplication & compression controller based, with compression module	Mandat ory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All- Flash Storage V7 Technical White Paper.pdf.pdf <u>Inline compression / Inline Deduplication / post-process deduplication:</u> Page no. - 61 - 65: Topic 5.1 SmartDedupe and SmartCompression (Data Reduction) , the proposed solution provides both inline and post-process deduplication and compression algorithms.		
18	Data Encryption:	Support Data at Rest Encryption solution for all drives.	Mandat ory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All- Flash Storage V7 Technical White Paper.pdf.pdf <u>Data Encryption:</u> Page no. 127 - 129: Topic 6.9 HyperEncryption (Array Encryption) , the proposed solution supports Data at Rest		

					Encryption solution for all drives.		
19	Storage Virtualization :	Should support storage virtualization and containerization	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf <u>Storage virtualization:</u> Page no. 72 - 74: Topic 5.3 SmartVirtualization (Heterogeneous Virtualization), the proposed solution supports SmartVirtualization storage feature for virtualization, <u>Containerization:</u> Page no. 61: The proposed solution supports SmartContainer (built-in container).		
20	Storage Features:	Thin Provisioning, volume copy, Compression and Deduplication, SAN to SAN FC over IP, FC and IP based Sync/Async local/remote replication	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf <u>Thin Provisioning:</u> Page no. 77: Topic 5.6 SmartThin (Intelligent Thin Provisioning), the proposed solution OceanStor Dorado supports thin provisioning, which enables the storage systems to allocate storage resources on demand. SmartThin does not allocate all capacity in advance, but presents a virtual		



					storage capacity larger than the physical storage capacity. <u>Asynchronous data replication:</u> Page no. 104 - 110: Topic 6.4 HyperReplication (Remote Replication), the proposed solution provides synchronous or asynchronous data replication, supporting intra-city and remote disaster recovery solutions.		
21		Immutable snapshot and secure replication in case of Cyber Threat / equivalent from same OEM	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf <u>Secure SnapShot:</u> Page no. 123 - 126: Topic 6.2 HyperCDP (Continuous Data Protection), the proposed solution creates secure snapshots.		
22		Storage should provide 100% data availability Guarantee in case of failure	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage Systems V7 Data Sheet.pdf <u>99.9999% Availability:</u> Page no. 3: Topic Ever Solid, the proposed solution supports delivers up to 99.999999% reliability for a single storage system thanks to improvements in components, architectures, systems,		

					and the storage solutions as a whole.		
23	Integration & Support:	Should support integration and provisioning from VMWare, Redhat Openshift, Openstack, etc, necessary drivers for containers (e.g CSI Drivers), file storage etc to be included	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf Integration: Page no. 188 - 189: Topic 13.2 Management and Control Plane Ecosystem Compatibility, the proposed solution provides support integration and provisioning from VMWare, Redhat Openshift, Openstack, etc, necessary drivers for containers (e.g CSI Drivers)		
24		Syslog integration with SIEM solution	Mandatory	FC	Reference to the supporting document Dorado 6000 V7 - SIEM with syslog, the proposed solution provides Syslog integration with SIEM solution.		
25		MFA on administrative login should be provisioned (MFA for 10 users with necessary integrations with soft tokens till duration of support period)	10	FC	Multi Factor Authentication (MFA): Reference to supporting document Dorado 6000 V7 - Multi-Factor Authentication.pdf, the proposed solution provides Multi Factor Authentication.	10	10



26	Storage Resource Pooling:	Should have ability to expand capacity or reconfiguration of LUNs, without downtime and outage.	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf <u>Storage Resource Pool:</u> Page no. 71: Topic 5.6 SmartThin (Intelligent Thin Provisioning), the proposed solution OceanStor Dorado supports thin provisioning, which enables the storage systems to allocate storage resources on demand. SmartThin does not allocate all capacity in advance, but presents a virtual storage capacity larger than the physical storage capacity.		
27	Quality of Service:	Proposed storages / solution should support Quality of Service (QoS) for prioritization of storage resources/services	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf <u>QoS:</u> Page no. 66 - 72: Topic 5.2 SmartQoS (Intelligent Quality of Service Control), the proposed solution supports SmartQoS that dynamically allocates storage system resources to meet the performance objectives of applications.		

28	Monitoring & Management Software:	Should provide management software of the proposed system, with web-based GUI interface capable of providing real time monitoring and able to generate customized reports from real time and historical performance data for analysis, trending and capacity utilization trends.	Mandatory	FC	Reference to the document New-Gen OceanStor Dorado 6000 Converged All-Flash Storage V7 Technical White Paper.pdf.pdf Intelligent GUI: Page no. 154 - 170: Topic 10.1.1 DeviceManager, the proposed solution provides Intelligent GUI, performance prediction, capacity prediction, Scheduled Reports.		
29	Licenses:	Should provide all required license (perpetual) of proposed Solution of regardless of capacity delivered capacity, no additional licenses should be required for any upgrade (storage virtualization, deduplication, compression, thin provisioning, cloning, replication (Sync/Async), snapshot, management and monitoring software's) with 03 years support and free of cost software renewals & upgrades.	Mandatory	FC	Reference to technical BOQ the offered storage array is equipped with SAN license bundle, and 03 year support software renewals & upgrades.		
30	Support and Warranty:	Should provide Three (03) years support 24 x 7 with NBD parts replacement and warranty services, Should provide future software / firmware releases free of cost and updates / upgrades them as recommended by the OEM during the 3 years period for the procured solution.	Mandatory	FC	Reference to technical BOQ the offered storage array is with 3 years support, 24 x 7 x 365 next business day.		

31	Product Life Cycle:	Proposed solution should not be declared End of Life and Support for at least 5 years from Acceptance.	Mandatory	FC	RoadMap Letter Attached.		
32	IOPS:	IOPS need to be shared by OEM certified document, actual lops of the proposed solution based on, I/O block size=8K Byte, RAID=6, Read%=60, write%=40, and need to be maintained in support period.	10	FC	IOPS: Reference to the technical BOQ, the propose storage solution provides 671672 IOPS .	10	10
33	Data Transfer M / B:	Data transfer rate of the proposed solution based on ,I/O block size=8K, RAID=6, Read%=60, write%=40.	10	FC	Data Transfer: Reference to the technical BOQ, the proposed storage solution provides Data Transfer Rate of 5247 MB/s .	10	10
					Obtained Marks =	85	100

ANNEX-B3: BACKUP STORAGE
(Quantity = 01)

S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/NC)	Quoted Feature Reference in Datasheet/ Remarks	Obtained Marks	Total Marks
1	Architecture:	Modular Unified Hybrid Storage System Backup Storage (Archive, backup/ cold data Storage)	Mandatory	FC	Reference to the document Huawei New-Gen OceanStor 5210 Hybrid Flash Storage Systems V7 Data Sheet.pdf Backup Storage: Page no. 1: The proposed solution Huawei OceanStor 5310 Hybrid Flash support hybrid workloads such as blocks, files, virtualization, and		



					containers, which meets the users' elastic service development requirements, improves storage resource utilization, and effectively reduces the total cost of ownership (TCO). In addition, balanced SAN and NAS services, supercharged by the Hyper and Smart features, provide diversified data protection and efficiency improvement capabilities for block and file systems. This provides users with comprehensive services.		
2		Active/Active Dual Controller High Availability (HA) Pair	Mandatory	FC	Reference to the document Huawei New-Gen OceanStor 5210 Hybrid Flash Storage Systems V7 Data Sheet.pdf Active Active: Page no. 1: The proposed solution is Industry's only gateway-free active-active solution for both SAN and NAS reduces the number of faulty nodes, simplifies deployment, and improves system reliability.		
3		Controller chassis form factor: 2U with >=12 internal drives	Mandatory	FC	Reference to the supporting document OceanStor 5210 Hybrid Flash - Controller Enclosure.pdf Controller: The proposed solution		

					supports 2 U controller enclosure with integrated disks: Supports 25 x 2.5-inch disks (same disk type as that in a 2 U 25-slot disk enclosure) Supports 12 x 3.5-inch disks (same disk type as that in a 4 U 24-slot disk enclosure)		
4		Drive Scalability: >=100 drives (Must supports SSD, NL-SAS, SAS, and mixed drive types) per HA pair	Mandatory	FC	Disk Scalability: Reference to the supporting document OceanStor 5210 Hybrid Flash - Maximum Disks.pdf, the proposed solution supports maximum 1000 disks per enclosure.		
5	Disk Type:	NL-SAS drives for RAID/Data	Mandatory	FC	Reference to the Technical BOQ the proposed solution is offered with 10 TB NL-SAS drives.		
6	Storage Capacity Preference:	Higher Storage capacity, better disk capacity type will be preferred.	10	FC	The proposed solution satisfies the storage capacity preference of Usable Capacity greater then 50 TB .	10	10
7	Hot-Spare Disk:	Additional 2 x NL-SAS drives, excluding data/RAID drives, more Drives will be preferred.	10	FC	Reference to the Technical BOQ the proposed solution offers Hot Spare Policy of 2 x NL-SAS drives .	10	10
8	Cache:	48GB memory per Controller	Mandatory	FC	Reference to the Technical BOQ the proposed solution offers 128 GB cache .		
9	Cache (Preference):	Higher than 48GB Memory will be preferred	20	FC	The proposed solution satisfies the cache storage perference of cache memory greater than 48GB .	20	2



10	RAID Configuration:	RAID 6, Dual and Triple parity RAID support	Mandatory	FC	Reference to the document Huawei New-Gen OceanStor 5210 Hybrid Flash Storage Systems V7 Data Sheet.pdf RAID: Page no. 3: The proposed solution supports RAID 10*, RAID 5, RAID 6, and RAID-TP (tolerating simultaneous failure of 3 disks).		
11	RAID Configuration (Preference):	Higher supported RAIDs will be preferred	5	FC	The proposed solution supports RAID 10*, RAID 5, RAID 6, and RAID-TP (tolerating simultaneous failure of 3 disks).	5	5
12	Backend Connectivity:	Atleast 12Gbps backend connectivity	Mandatory	FC	The proposed solution is equipped with backend connectivity.		
13	Frontend Ports:	2 x 32 Gbps optical interfaces per controller	Mandatory	FC	Frontend Ports: Reference to the technical BOQ, the proposed storage solution provides (2 x 32 Gbps optical interfaces) per controller.		
14		Or 4 x 25 Gbps iSCSI optical interfaces per controller	Mandatory	FC	Frontend Ports: Reference to the technical BOQ, the proposed storage solution provides (2 x 10 Gbps) iSCSI optical interfaces per controller.		
15	Frontend Ports (Preference):	More/Higher ports will be preferred	5	FC	Reference to the Technical BOQ the proposed solution provides higher ports for better connectivity and compatibility.	5	5



16	Protocol Support:	Protocols and license as per provided solution interfaces, higher/better protocols will be preferred.	5	FC	Reference to the document Huawei New-Gen OceanStor 5210 Hybrid Flash Storage Systems V7 Data Sheet.pdf Protocols: Page no. 3: The proposed solution supports FC, iSCSI, NFS, CIFS, FC-NVMe, NFS over RDMA, NVMe over RoCE, FTP*, NDMP Front-End Storage Protocols.	5	5
17	Preference Encryption, Deduplication / Compression:	Higher values of compression, deduplication, Worm-in encryption will be preferred	5	FC	Compression: Reference to the supporting document OceanStor 5210 Hybrid Flash - Compression.pdf, the proposed solution provides Compression granularity of data blocks (8 KB/16 KB/32 KB). Deduplication: Referene to the supporting document OceanStor 5210 Hybrid Flash - Deduplication.pdf, the proposed solution provides Deduplication granularity of data blocks (8 KB/16 KB/32 KB).	5	5
18	Storage Virtualization / Containerization:	Must support storage virtual machines, and Storage Containerization services (necessary drivers / software to be included)	5	FC	Reference to the document OceanStor 5210 Hybrid Flash Storage V7 Technical White Paper.pdf Virtualization: Page no. 46: Toipc 5.3 SmartVirtualization	5	

					(Heterogeneous Virtualization), the proposed solution uses SmartVirtualization to take over LUNs from heterogeneous storage systems. In addition, SmartVirtualization can work with SmartMigration to migrate data from heterogeneous storage systems online, facilitating device replacement. Containerization: Page no. 156: Topic 12.2.4 Container Platform Integration, the proposed solution OceanStor hybrid flash storage releases CSI Plugin and FlexVolume Plugin, and supports mainstream container management platforms such as Kubernetes and Red Hat OpenShift. The OceanStor hybrid flash storage releases the CDR plugin to provide backup and recovery capabilities for containerized applications, ensuring data security of mission critical services.		
19	Storage Features:	Anti-ransomware built-in protection for both preemption and post-attack recovery. Logically air gapped and	5	FC	Reference to the document Huawei New-Gen OceanStor 5210 Hybrid Flash Storage Systems V7 Data Sheet.pdf Ransomware Protection: Page no. 1:	5	

		Immutable Snapshot.			Toipc Endless Evolution, the proposed solution supports SAN + NAS ransomware protection solution that features the Air Gap technology and high-density snapshots can effectively defend against ransomware attacks and support fine-grained data restoration*. Reference to the document OceanStor 5210 Hybrid Flash Storage V7 Technical White Paper.pdf <u>Secure Snapshot:</u> Page no. 67 - 69: Topic 6.2 HyperCDP (Continuous Data Protection) sub topic Secure Snapshot, the proposed solution create schedule policies for secure snapshots.		
20		Including Multi-factor authentication (out of band 2FA), and role-based access control, with all required hardware/software.	5	FC	<u>RBAC:</u> Reference to the supporting document OceanStor 5210 Hybrid Flash - RBAC.pdf The proposed solution provides role based access control, ensuring security of the storage system and service data. <u>MFA:</u> Reference to the supporting document OceanStor 5210 Hybrid Flash - Multi Factor Authentication.pdf,	5	5



					the proposed solution provides multi factor authentication capability.		
21		Operating support for all leading operating systems (including but not limited to M/S Windows, RHEL, OEL etc) and virtualization (including but not limited to Vmware, Redhat, OpenStack and Other industry standard virtualization/Cloud solutions and others). Preference will be given to more integratable solution	5	FC	Reference to the document OceanStor 5210 Hybrid Flash Storage V7 Technical White Paper.pdf <u>Operating System:</u> Page no. 154: Topic 12.1.1 Host Operating System, the proposed solution supports the following operating systems. - IBM AIX - HP-UX - Solaris - Red Hat Enterprise Linux - SuSE Linux - Enterprise Server - Oracle Enterprise Linux - Windows Server - NeoKylin, Kylin (Tianjin) - Hunan Kylin - Deepin - Linx-TECH Rocky - Redflag Linux <u>Virtualization:</u> Page no. 154 - 155 : Topic 12.1.2 Host Virtualization System, the proposed solution supports the following Virtualization platforms. - VMware ESXi - Microsoft Hyper-V - XenServer - Red Hat RHV - IBM PowerVM (VIOS) - Huawei FusionCompute - NeoKylin advanced server operating	5	5



					system (virtualization version)		
22	Storage Resource Pooling:	Should have ability to expand capacity or reconfiguration of LUNs, without downtime and outage.	Mandatory	FC	Reference to the document OceanStor 5210 Hybrid Flash Storage V7 Technical White Paper.pdf Storage Resource Pooling: Page no. 50 - 51: Topic 5.6 SmartThin (Intelligent Thin Provisioning), the proposed solution triggers storage resource pool expansion to add more space.		
23	Quality of Service:	Proposed storages / solution should support Quality of Service (QoS) for prioritization of storage resources/service s.	5	FC	Quality of Service (QoS): Reference to the supporting document OceanStor 5210 Hybrid Flash - Quality of Service.pdf, the proposed solution supports: 1024 (32 GB or lower memory per controller) 2048 (64 GB memory per controller)	5	5
24	Monitoring & Management Software:	Must have built in storage managemnet software for Operational simplicity with a streamlined setup process and intuitive management through the GUI, CLI and REST APIs.	Mandatory	FC	Reference to the document OceanStor 5210 Hybrid Flash Storage V7 Technical White Paper.pdf CLI: Page no. 137: Topic 9.1.2 CLI, the proposed solution provides Command Line Interface (CLI) allows administrators and other system users to manage and maintain the storage system. It is based on the secure shell		



					protocol (SSH) and supports key-based SSH access. REST APIs: Page no. 137: Topic 9.1.3 RESTful APIs, the proposed solution provides RESTful APIs. RESTful APIs of the OceanStor hybrid flash storage allow system automation, development, query, and allocation based on HTTPS interfaces. With RESTful APIs, you can use third-party applications to control and manage arrays and develop flexible management solutions for the OceanStor hybrid flash storage.		
25	Licenses:	Full proposed capacity licences must be included	Mandatory	FC	Reference to technical BOQ the offered storage array is equipped with all the required license bundle.		
26	Licenses (Preference):	Preference will be given to product with more licenses and features.	5	FC	Reference to technical BOQ the offered storage array meets the requirement for licenses and features.	5	5
27	Support and Warranty:	3x years OEM 24/7/365, NBD part replacement support with license renewal, security patches and Firmware upgrade services	Mandatory	FC	Reference to technical BOQ the offered storage array is equipped with 03 year support software renewals & upgrades.		
28	Product Life Cycle:	The proposed storage solution should not reach End of Sale (EoS) for at least five	Mandatory	FC	RoadMap Letter Attached.		



		years from the date of sale.					
29	IOPS:	IOPS need to be shared by OEM certified document, actual lops of the proposed solution based on, I/O block size=8K Byte, RAID=6, Read%=60, write%=40, and need to be maintained in support period.	5	FC	IOPS: Refer to the sizer document.	5	5
30	Data Transfer M / B:	Data transfer rate of the proposed solution based on ,I/O block size=8K, RAID=6, Read%=60, write%=40.	5	FC	Data Transfer: Refer to the sizer document.	5	5
Obtained Marks =						100	100

ANNEX-B4: SAN SWITCHES					
S#	Items	Required Item Description	Compliance	Bidder Compliance (FC/PC/NC)	Quoted Feature Reference in Datasheet/ Remarks
1	Quantity	Number of (02) Gen 7 SAN switches.	M	FC	Lenovo Gen 7 ThinkSystem DB720S SAN switch
2	Port per Switch	24 ports activated with 32Gbps SFP (with support for 32Gbps, 64Gbps) and all required license having key features i.e. NPIV, SFPs for all Ports.	M	FC	24 Ports Active with 32Gb SWL SFPs and refer to Datasheet
3	ISL connectivity & management	Should provide required ISL license for all switches, Should provide Ethernet / Serial port for management	M	FC	24 Ports Active/ licenced with 32Gb license refer to Datasheet
4	High Availability	Dual Redundant Power supplies for Aile containment (as per solution), and all components need to be redundant (e.g connectivity, Fan etc)	M	FC	Dual, hot-swappable redundar power supplies with integrater system cooling fans refer to datasheet



5	Management & Monitoring	a. Should provide web management through http / https and command line management through SSH	M	FC	refer to datasheet
6		b. Should provide real time utilization, health status and performance bottleneck in SAN.	M	FC	Visualize application and device-based performance and health metrics
7		c. Should provide role based user access on switch for each user.	M	FC	user-defined Role-Based Access Control (RBAC)
8	Licenses	Should provide licenses for all functionality and software offered with this solution with free of cost	M	FC	24 Ports Active/ licenced with 32Gb license refer to Datasheet
9	Compatibility	switches firmware should be certified with proposed storage system and existing infrastructure	M	FC	3-year hardware and firmware/FOS (upgrades available)
10	Cables & accessories	Should provide rack mounting kit and power cables for installation of switches.	M	FC	rail kit and power cables included
11	Support & Warranty	a. Should provide Three (03) years 24 x 7 NBD warranty with subscriptions, parts, labour and support.	M	FC	Refer to BoQ
12		b. Should provide future software / firmware releases free of cost and updates, renewals, upgrades them as recommended by the OEM	M	FC	3-year hardware and firmware/FOS (upgrades available)
13	Product Life Cycle	Proposed solution should not be declared End of Life and Support for at least 5 years from Acceptance.	M	FC	

ANNEX-B5: CLOUD AND VIRTUALIZATION

Sr.	Component & Scope (Subscription)	Compliance	Bidder's Statement	
			(PC/FC/NC)	Quoted Feature Reference in Datasheet
1	Cloud Infrastructure Virtualization (IaaS) The solution needs to be on-premises self-hosted Cloud in multitenancy. Solution needs to be in Cluster (high availability), nodes for cloud management (separate management nodes if required for deployment of solution to be quoted in addition to workload nodes)	Mandatory	FC	Refer to NexaVM - Technical Whitepaper
i	Hypervisor/ Virtualization hosts for Virtual Machines (VMs). Must support standard virtualization features, including compute, storage, Network (internet and	Mandatory	FC	Refer to NexaVM Cloud - Technical Whitepaper

	intranet services), Load balancing, auto-scaling, VPN, Basic Backup utility (including snapshot based backup), Monitoring and other network virtualization features in multitenancy to enable NTC provide services in IaaS Model. Enterprise Kubernetes solution includes all standard features of Kubernetes for Container services, Backup/ Monitoring/ Logging etc. Solution needs to support restful & open API. Solution will include IAC (Infrastructure as a Code), and support Of DevOps mechanism using scripting, ansible, terraform etc.			
ii	Backup: Application Aware Backup, capable of integrating with hypervisor, for entire solution (on all server processors i.e 12), solution to support backup life-cycle policies, including immutability and encryption of backup to ensure security of the backup data. Must Support integrated Disaster Recovery functionality in the solution.	Mandatory	FC	Refer to NexaVM nBDR - Brochure
lii	GPU Virtualization License (for all compute machines where applicable to be offered with cloud offering as a solution. Solution needs to be integrated with GPU, hardware for the project.	Mandatory	FC	Refer to BOQ
2	Unified Orchestration capable of managing multitenant heterogeneous environments.	Mandatory	FC	Refer to NexaVM Cloud - Technical Whitepaper
i	Unified Management in multitenancy, and necessary access controls including IAM & Security (Identity and Access Management (IAM). Must deploy and integrate with Directory Services (LDAP/ AD) as per solution requirement.	Mandatory	FC	Refer to NexaVM Cloud - Feature List
ii	Metering (Built-in module for metering a(online/offline) supporting various service models.	Mandatory	FC	Refer to NexaVM Cloud - Feature List
3	Professional Services & Deployment & SLA (3 x years SLA including all licenses validity effective from PAC of project) i.e. One Year Standard Warranty and Two Years SLA.	Mandatory	FC	Refer to BOQ
i	OPTIONAL: Year wise support for Year 4 & 5	Mandatory	FC	Refer to BOQ
ii	Deployment, Design & Validation: Conduct design workshops, create Low-Level Design (LLD) documentation, perform pre-deployment environment validation, Deploy the IaaS, including necessary repositories (Image Registries/ Artifact Repositories), complete solution deployment capable of onboard services through Cloud setup. Provide tenants with onboarding documentation and operational handover.	Mandatory	FC	Refer to BOQ

